Supplementary Information

SAMPA: Segment Anything Model for Plant Image Analysis

Narendra Narisetti[1*] and Evgeny Gladilin[1]

[1: Leibniz Institute of Plant Genetics and Crop Research, 06466, Seeland, OT Gatersleben,

*: Corresponding author email- narisetti@ipk-gatersleben.de]

S1. Phenotypic Trait Extraction

Root System Module. The root trait computation pipeline applies the following sequence to each binary label mask:

Projected area is the sum of foreground pixels. Root length is computed using a binary skeleton generated by medial-axis thinning (Zhang–Suen algorithm [15]) followed by summation of one-pixel-wide skeleton. Diameter at each skeleton pixel is estimated from the distance transform of the mask, defined as twice the distance from each skeleton pixel to the nearest background pixel. Surface area and volume are computed under a cylindrical approximation by integrating the local diameter along the skeleton: SurfaceArea = $\Sigma(\pi \times d_i \times s_i)$ and Volume = $\Sigma(\pi \times (d_i/2)^2 \times s_i)$, where d_i is the local diameter and s_i is the length element at skeleton pixel i. Convex hull metrics (CH_width, CH_height, CH_area) are derived from the convex hull polygon of the mask pixel coordinates using MATLAB's convhull function. Specific convex hull area (CH_area_specific) is the ratio of CH_area to projected area, providing a measure of how completely the root fills its occupied space. Spatial distribution statistics (Mean_X, Stdev_X, Mean_Y, Stdev_Y) are the mean and standard deviation of the x- and y-coordinates of all foreground pixels. Horizontal position percentiles (X_p25, X_p50, X_p75, X_p90, X_p95, X_p99) are the 25th through 99th percentiles of the x-coordinate distribution, quantifying lateral root spread at fine resolution. The full trait set is summarised in Table 1.

Table S1: Phenotypic traits computed by SAMPA for the Root and Shoot System image type.

Trait	Column	Computation	Units
Projected area	Area / Shoot_area	Count of foreground pixels	px2
Root length	Length	Medial-axis skeleton, summed all skeleton pixels	px
Estimated surface area	SurfaceArea	$\Sigma (\pi \times d_i \times s_i)$ over skeleton	px2
Estimated volume	Volume	$\Sigma (\pi \times (d_i/2)^2 \times s_i)$ over skeleton	px3
Convex hull width	CH_width	Bounding box width of convex hull	px
Convex hull height	CH_height	Bounding box height of convex hull	px
Convex hull area	CH_area	Polygon area of convex hull	px2
Specific convex hull area	CH_area_specific	CH_area / Area	--
Mean horizontal or vertical	Mean_X or Mean_Y	Mean x or y-coordinate of mask pixels	px

position			
SD horizontal or vertical position	Stdev_X or Stdev_Y	Standard deviation of x or y-coordinates	px
Percentile positions	X_p25 … X_p99 or Y_p25 … Y_p99	Percentiles of x or y-coordinate distribution	px
Bounding box area	BBX_area	Area of the rectangle bounding box	px2
Shoot area to bounding box area	Shoot_area to BBX_area	Shoot_area / BBX_area	--
Shoot Bounding box height	BBX_height	Height of the rectangle bounding box	px
Shoot Bounding box width	BBX_width	Width of the rectangle bounding box	px
Shoot Convex hull area to bounding box area	CH_area to BBX_area	CH_area / BBX Area	--
Shoot area to convex hull area	Shoot_area to CH_area	Shoot_area / CH_area	--
Mean colour pixels of shoot	R_mean, G_mean, B_mean, H_mean, S_mean, V_mean, L_mean, a_mean, b_mean, hny_H_mean, hny_S_mean, hny_Y_mean	Mean values of RGB, HSV, Lab and HSY (hny) colour spaces	--
SD colour pixels of shoot	R_stdev, G_ stdev, B_ stdev, H_ stdev, S_ stdev, V_ stdev, L_ stdev, a_ stdev, b_ stdev, hny_ stdev, hny_S_ stdev, hny_Y_ stdev	Standard deviation values of RGB, HSV, Lab and HSY (hny) colour spaces	--

Shoot System Module: For shoot system images, SAMPA applies a complementary trait set: projected leaf/canopy area, plant height (bounding box height of mask), plant width (bounding box width), solidity (Area / CH_area), and the colour (RGB, HSV, Lab, and HSY) related statistics as proxies for chlorophyll content and vigour.